SE Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #07: Hostel Maintenance & Issue Ticketing System

DOMAIN:-Campus & Academic Operations

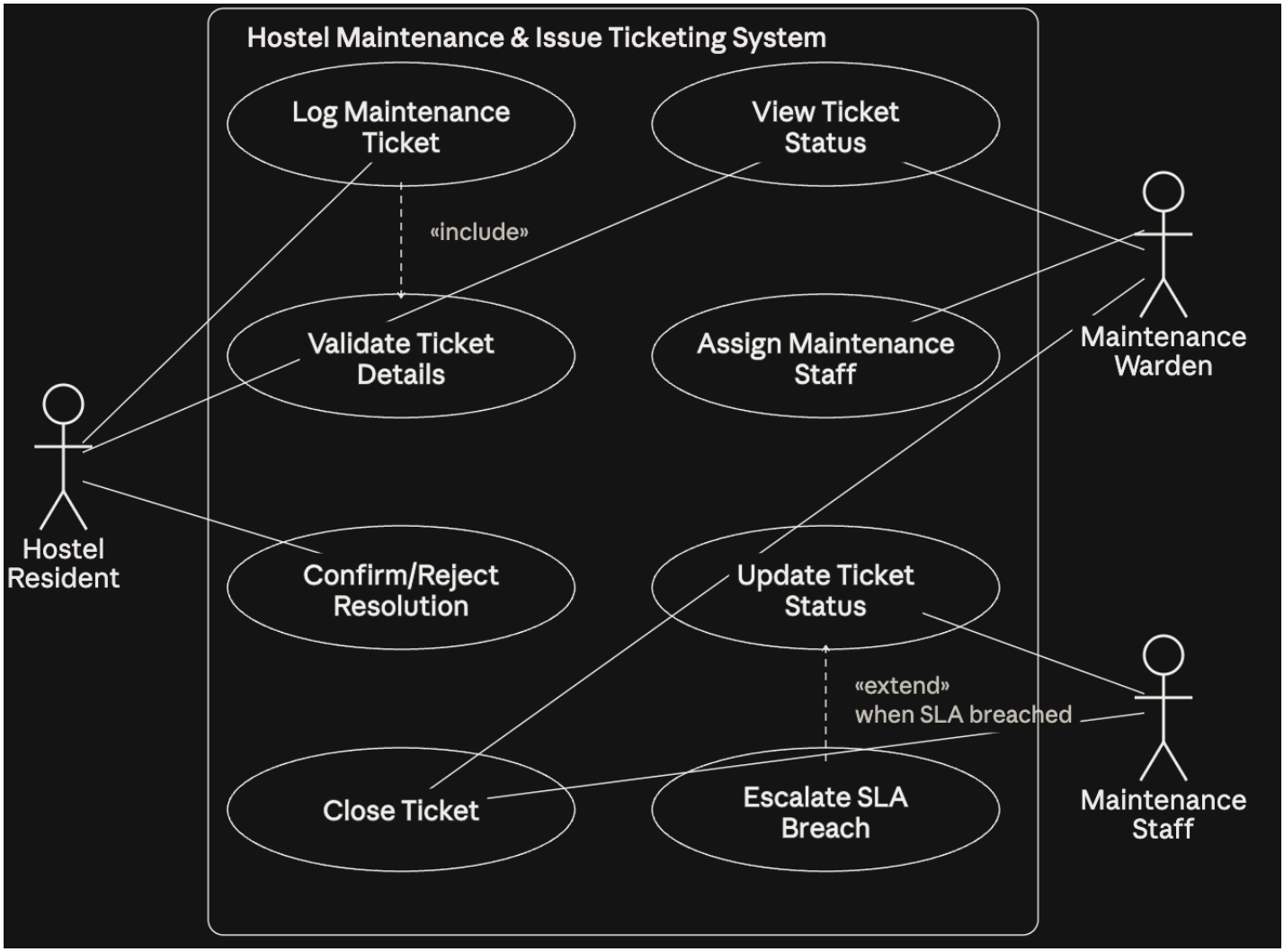
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1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable hostel residents to log maintenance tickets with a room number tag, photo attachment, issue category, and severity level.	High	Pass: A complete ticket is created with a unique ticket ID and appears in the warden queue. Fail: A ticket is accepted without a room number tag or required issue details.	Provides the core mechanism for reporting plumbing, electrical, and carpentry issues.
FR-002	Functional	The system shall allow the Maintenance Warden to review incoming tickets and assign each ticket to an available maintenance staff member with a recorded assignment status.	High	Pass: The selected ticket shows the assigned staff member and assignment timestamp. Fail: The system allows a ticket to be marked assigned without a staff member.	Enables dispatch of maintenance staff and makes responsibility for each ticket traceable.
FR-003	Functional	The system shall allow residents and the Maintenance Warden to view the current status, priority, assigned staff member, and SLA deadline of a maintenance ticket.	High	Pass: For every valid ticket ID, the system displays status, priority, assignee, and SLA deadline. Fail: Any of these fields is missing for an active ticket.	Provides visibility into ticket progress and SLA compliance.
FR-004	Functional	The system shall allow the Maintenance Warden to update a ticket status through defined states such as Open, Assigned, In Progress, Resolved, and Closed, and shall record the time of each status change.	High	Pass: A status change is saved and its timestamp is recorded in the ticket history. Fail: An invalid status transition or an update without a timestamp is stored.	Supports end-to-end tracking of maintenance work and creates an auditable history.
FR-005	Functional	The system shall allow a resident to confirm or reject a resolution for their ticket, and shall close the ticket only after resident confirmation or an authorized warden closure action.	High	Pass: A confirmed resolution changes the ticket to Closed and records the resident sign-off. Fail: A ticket is closed without a confirmation/sign-off or authorized warden action.	Ensures that issue closure reflects resident sign-off and satisfies the scenario's closure requirement.
NFR-001	Nonfunctional – Performance & Security	The system shall trigger automated escalation SMS/email alerts if a high-priority ticket exceeds a 24-hour SLA without a status change.	High	Pass: In a simulated test, a high-priority ticket with no status change for 24 hours generates the configured escalation alert. Fail: No alert is generated, or an alert is generated before the 24-hour threshold.	Prevents high-priority maintenance issues from remaining unattended and follows the given SLA guideline.
NFR-002	Nonfunctional – Availability & Data Integrity	The system shall preserve ticket details, status history, attachments, assignment information, and closure sign-off so that an authorized user can retrieve the complete record for every ticket.	Medium	Pass: After a save and subsequent retrieval, all required ticket fields, history entries, attachments, and sign-off data are present and unchanged. Fail: Any required record is lost, corrupted, or inaccessible to an authorized user.	Maintains reliable maintenance records for tracking, accountability, and future reference.

2.



Actors-Hostel residence , maintenance warden and maintenance staff

Relationship :- <<include>> and <<extend>>

3. *Use-Case Flow Specification*

Preconditions

- The Hostel Resident is authenticated in the Hostel Maintenance & Issue Ticketing System.
- The resident has a valid hostel room number associated with their account.
- The ticketing service is available.

Postconditions

- A maintenance ticket is stored with a unique ticket ID.
- The ticket contains the room number, category, severity, description, and any supplied photo attachment.
- The ticket is routed to the Maintenance Warden queue with an initial Open status.
- The ticket's SLA information is recorded.

Main Success Scenario

1. Resident selects “Log Maintenance Ticket”.

2. System displays the ticket form.

3. Resident enters/selects room number, issue category, issue description, and severity, and optionally attaches a photo.

4. System validates the required fields and attachment format.

5. System creates a unique ticket ID and records the ticket.

6. System calculates/records the applicable SLA deadline based on ticket priority.

7. System routes the ticket to the Maintenance Warden queue.

8. System displays the ticket ID and current Open status to the resident.

9. Use case ends successfully.

Alternate Flow

4a. Missing or invalid required information: The system highlights the invalid fields and displays a message explaining what must be corrected.

4a1. Resident corrects the information and resubmits the form.

4a2. The system validates the corrected data and resumes at Step 5.

4b. Invalid photo attachment: The system rejects the attachment, displays the allowed format/size requirement, and allows the resident to continue without the invalid attachment or select a valid one.